

A Prefetchers

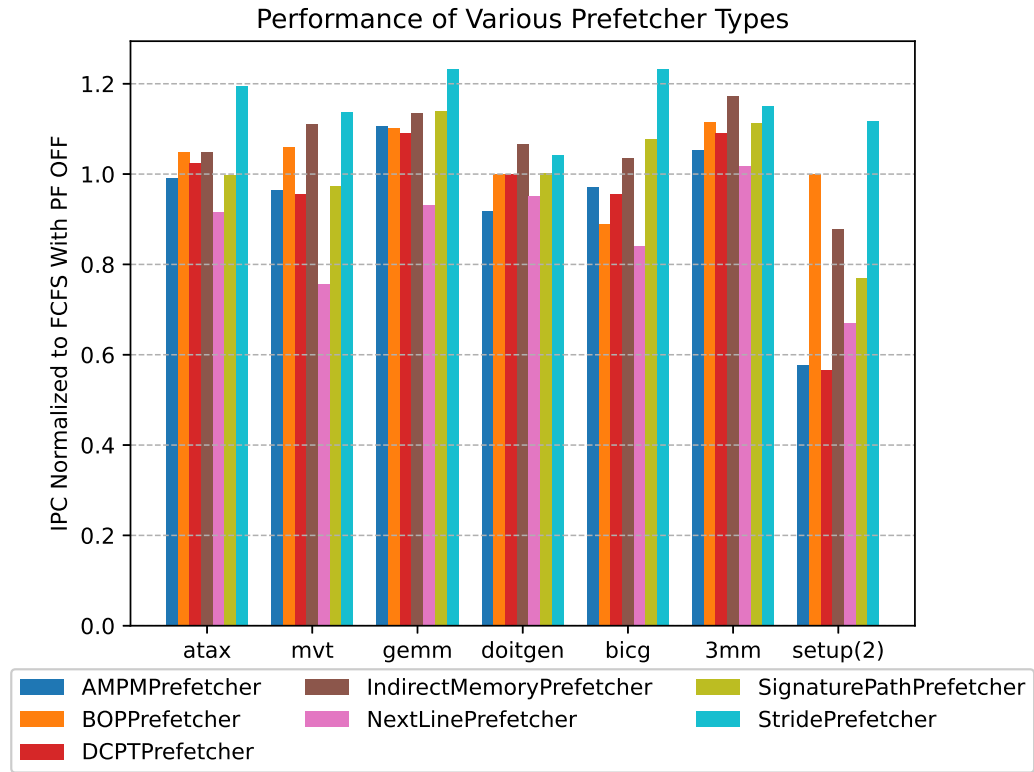


Figure 12 Normalized IPC for various prefetcher types running benchmarks under setup 1 and setup 2

Figure 12 presents the average performance of different prefetcher types across various PolyBench benchmarks under setup 1 and setup 2. These experiments adopt the predictable prefetcher setup discussed in Section 4, where each prefetcher includes a prefetch side buffer with eight entries. The stride prefetcher outperforms all others in every benchmark except *doitgen*. The figure also shows that under setup 2, the AMPM, DCPT, IndirectMemory, NextLine, and SignaturePath prefetchers perform poorly. This can be attributed to the fact that this set of PolyBench benchmarks is particularly suited to constant stride patterns only. Given the stride prefetcher’s effectiveness and simple hardware mechanism for detecting such patterns, we selected it for our experiments.